Gaurav Singh

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Summary

Computer Science student skilled in Backend Development, System Design, Cloud Computing (AWS), AI/ML, and Data Structures & Algorithms.

Skills

Programming Languages: C++, Python, JavaScript, SQL, HTML, CSS, Objective-C

Backend Development: Node.js, Express.js, FastAPI, REST APIs, WebSockets, Authentication & Authorization

Frontend Development: React.js, Next.js, Tailwind CSS

Databases: MongoDB, MySQL, Firebase Firestore, PostgreSQL

Cloud & DevOps: AWS (Cloud Practitioner), Docker, Kubernetes, CI/CD

System Design & Architecture: System Design, Microservices, Load Balancing, API Gateway

Developer Tools: Git, GitHub, Postman, Clerk, Render, Vercel, Netlify

AI Tools: Claude Opus 4, Gpt 5, Cursor AI, NotebookLM, Google Stitch, V0, Drizzle ORM, Vercel, LLM Integration

Core Computer Science: Data Structures & Algorithms, Object-Oriented Programming (OOP), Database Management Systems (DBMS), Operating Systems (OS), Computer Networks (CN), Natural Language Processing (NLP)

Soft Skill: Team Collaboration, Communication, Responsibility, Adaptability, Time Management

Experience

ANRF (SERB) — TARE Project Intern, Department of Science & Technology, Govt. of India
- Building a real-time vehicle tracking system using YOLOv8, OpenCV, and FastAPI to detect and track vehicles from aerial footage.

Technical Team Member — IEEE SRMIST
- Contributing to web development and tech infrastructure for IEEE SRMIST, supporting technical events for 800+ students.

Projects :

CareerCompass - AI-Powered Career Guidance Platform - (React, Node.js, MongoDB, LLM) [View Project]
- Developed an AI-powered career exploration platform for 10th and 12th-grade students, featuring 50+ career pathways, LLM-based personalized recommendations, interactive career roadmaps, advanced search functionality, and Maps API integration for discovering nearby colleges and educational institutions.

A FiT Model based Early Diagnosis Framework for Ocular Disorders
- Developed an AI-based early diagnosis framework using FiT models for ocular disorder detection from medical images.

Flooded Area Estimation Using DeepLab V3 and Drones Image
- Developed an end-to-end flood segmentation pipeline using DeepLabV3+ (ResNet-50) on drone imagery achieving 85.21% pixel accuracy, with GSD-based area estimation and A* rescue path planning.

Certifications

- AWS Certified Cloud Practitioner — [Certificate]
- Database Management System (NPTEL) — [Certificate]
- AWS Cloud Virtual Internship — [Certificate]
- Smart India Hackathon (SIH) — [Certificate]

Research & Publications

A FiT Model-based Early Diagnosis Framework for Ocular Disorders
International Conference on Communication, Computing and Emerging Technologies (IC3ET 2026 – IEEE)
Best Paper Award Certificate

Achievement

"WhereTheFlag 2.0" CTF, HackTheBox SRMIST [3rd Place]

ANRF (SERB) — TARE Project Intern [3rd Place]

Awarded by SRM Institute of Science and Technology for academic excellence [Performance Scholarship]

Certificate of Merit — SRMIST Science Exhibition and IoT Exhibition [1st Place]

Solved 450+ Data Structures and Algorithms problems across LeetCode, GeeksforGeeks, and other coding platforms.

Education

Bachelor of Technology in Computer Science [2023-2027]
CGPA: 9.65/10
SRM Institute of Science and Technology

Class 12 (Percentage: 86.4%)

Class 10 (Percentage: 82.9%)